

Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

Task details	
Task	cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).
Dataset	Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens.
Classes	Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).
Split	70/15/15 train/val/test, 523 test samples , seed 42.
Model	Recursive Genomic Q-Former: $d_{\text{model}} = 128$, 256 marker query slots, recursion depth 4.
Metric	Test accuracy and macro-averaged F1.

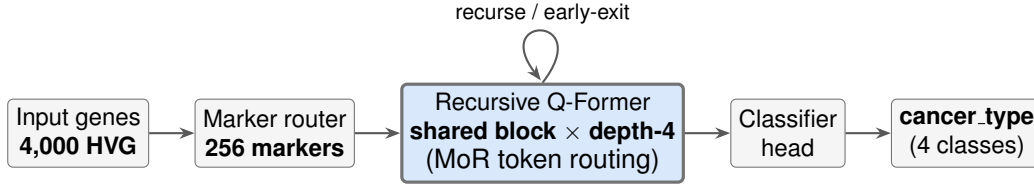


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4, $d_{\text{model}} = 128$, 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

Table 1: Parameters and performance for all three model types. One view of the full trade-off: parameter budget (recurrent-block params, total params, and the independent/shared ratio) alongside all three performance metrics (accuracy, macro-F1, weighted-F1). Sharing the transformer block cuts recurrent params by $\sim 4\times$ for a 0.57-point accuracy cost; Mixture-of-R recursions (MoR) keeps that small budget yet recovers the gap to 0.19 points and is best on macro-F1.

Model	Weights	Routing	Parameters			Performance		
			Recurrent	Total	Ratio	Acc.	Macro-F1	Wtd-F1
Independent stack	untied	none	530,432	1,109,897	$3.99\times$	0.9904	0.9891	0.9904
Recursive (shared)	shared	none	132,992	712,457	$1.00\times$	0.9847	0.9871	0.9847
MoR (token-adaptive)	shared	token	132,992	712,457	$1.00\times$	0.9885	0.9892	0.9885

Read-out. The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

Table 2: Parameter scaling vs. recursion depth. Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$ at depth 8.

Recursion depth	Shared params	Independent params	Reduction factor
1	132,608	132,608	1.00 $\times$
2	132,736	265,216	2.00 $\times$
4	132,992	530,432	3.99 $\times$
6	133,248	795,648	5.97 $\times$
8	133,504	1,060,864	7.95 $\times$

Table 3: Compute trade-off (depth-4 variants). Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

Configuration	Transf. params	Mean depth	MFLOPs/sample	Test acc.
Fixed depth-4 (no early exit)	132,480	4.00	268.4	0.9847
Recursive (shared, early-exit)	132,992	2.75	161.5	0.9847
No recursive refine	132,992	2.75	161.5	0.9771
MoR (token-adaptive)	132,992	2.69	159.1	0.9885

Notes. Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is ~ 712 K for shared variants, ~ 1.11 M for the independent stack). Compute-saving for the early-exit and MoR variants is $\approx 40\%$ of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.

Cohort-wise clinical prediction

The model is evaluated on the **unified TCGA benchmark**: 2,738 patient samples, 20,530 genes (top 4,000 highly-variable genes used as input tokens). Tumor-of-origin (`cancer_type`) is modeled *unified* across all four cohorts; the clinical and outcome targets are modeled *cohort-by-cohort* (breast, lung, thyroid, head&neck), so staging and survival are learned without cross-cohort confounding. Because the cohorts are small and class-imbalanced, all targets are evaluated with **stratified 5-fold cross-validation** and reported as **macro-F1 (mean $\pm$ std)** — accuracy on these splits collapses to the majority-class floor and is identical across models, which hides the true difficulty.

Table 4: Cohort-wise macro-F1 of the three model types (5-fold CV). `cancer_type` is modeled *unified* (pooled across all four cohorts, it is the cohort identity); the five clinical / outcome targets are modeled *separately within each cohort* (breast, lung, thyroid, head&neck), for all three model types. Macro-F1 (% , mean $\pm$ std over stratified 5-fold cross-validation) is reported instead of accuracy: on these small, imbalanced cohorts accuracy collapses to the majority-class floor (and is identical across models), whereas macro-F1 exposes the true difficulty. Best per row in **bold**; recurrent-block parameters under each model.

Cohort	Task	Macro-F1 % (mean $\pm$ std, 5-fold CV)		
		Independent 530,432 par.	Recursive (shared) 132,992 par.	MoR 132,992 par.
Unified (all)	<code>cancer_type</code>	97.9 $\pm$ 0.9	96.9 $\pm$ 0.4	96.7 $\pm$ 0.9
Breast (867)	<code>pathologic_stage</code>	26.4 $\pm$ 3.8	25.1 $\pm$ 3.9	26.6 $\pm$ 3.2
	<code>pathologic_T</code>	26.6 $\pm$ 4.4	26.6 $\pm$ 3.7	24.8 $\pm$ 2.3
	<code>pathologic_N</code>	22.8 $\pm$ 1.3	21.2 $\pm$ 2.6	20.7 $\pm$ 4.6
	<code>os_binary</code>	50.2 $\pm$ 5.7	50.4 $\pm$ 3.9	49.3 $\pm$ 5.9
	<code>tumor_status</code>	51.6 $\pm$ 3.4	47.4 $\pm$ 5.0	50.8 $\pm$ 3.0
Lung (943)	<code>pathologic_stage</code>	26.1 $\pm$ 2.5	24.2 $\pm$ 2.9	27.8 $\pm$ 1.6
	<code>pathologic_T</code>	27.3 $\pm$ 2.2	24.7 $\pm$ 1.6	22.4 $\pm$ 3.3
	<code>pathologic_N</code>	22.7 $\pm$ 2.0	24.1 $\pm$ 3.5	24.5 $\pm$ 2.5
	<code>os_binary</code>	46.1 $\pm$ 1.6	46.9 $\pm$ 0.9	46.8 $\pm$ 0.9
	<code>tumor_status</code>	55.0 $\pm$ 2.8	52.6 $\pm$ 5.2	53.1 $\pm$ 3.2
Thyroid (505)	<code>pathologic_stage</code>	25.3 $\pm$ 3.4	24.5 $\pm$ 4.0	24.6 $\pm$ 4.7
	<code>pathologic_T</code>	24.8 $\pm$ 5.9	24.5 $\pm$ 3.3	25.3 $\pm$ 2.4
	<code>pathologic_N</code>	58.0 $\pm$ 3.4	64.8 $\pm$ 2.7	58.9 $\pm$ 4.9
	<code>os_binary</code>	49.2 $\pm$ 2.2	48.7 $\pm$ 0.2	47.2 $\pm$ 3.0
	<code>tumor_status</code>	50.6 $\pm$ 7.3	47.8 $\pm$ 2.1	47.2 $\pm$ 3.2
Head&neck (423)	<code>pathologic_stage</code>	20.4 $\pm$ 4.9	21.8 $\pm$ 3.9	19.2 $\pm$ 2.3
	<code>pathologic_T</code>	20.8 $\pm$ 3.8	20.4 $\pm$ 3.6	19.5 $\pm$ 2.3
	<code>pathologic_N</code>	25.8 $\pm$ 3.8	24.9 $\pm$ 5.5	28.8 $\pm$ 3.9
	<code>os_binary</code>	47.5 $\pm$ 0.2	47.9 $\pm$ 1.6	48.7 $\pm$ 2.3
	<code>tumor_status</code>	50.8 $\pm$ 4.7	51.2 $\pm$ 4.0	50.0 $\pm$ 2.1

Notes. Macro-F1 %, mean $\pm$ std over stratified 5-fold cross-validation (every sample tested once), recurrent-block parameters under each header (total ~ 1.11 M Independent; ~ 712 K shared and MoR). `cancer_type` is the unified four-cohort model (`results_cohorts_cv/*/unified/`); the per-cohort clinical/outcome results are from `results_cohorts_cv/*/(<cohort>)/`. *Reading the table:* only tumor-of-origin (`cancer_type`, $\sim 97\%$) and node status in thyroid (`pathologic_N`, $\sim 59\text{--}65\%$) carry strong signal; the 4-class staging targets hover near the 25% chance level and the binary outcome targets near 50%, confirming that fine-grained clinical staging is not recoverable from bulk expression at these cohort sizes. No single model type dominates, and differences are mostly within one std.

Single-cell cell recognition (genomap benchmark)

For an apple-to-apple comparison with the genomap paper (Islam & Xing, *Nat. Commun.* 2023), we evaluate the same recursive model, in its three variants, on the genomap classification datasets under the **paper’s exact protocol**: precomputed genomap features (5–10% HVG + entropy cartography), the paper’s train/test split (Tabula Muris 70/30; pancreas integration split), Adam + weight decay (lr 0.001, wd 10^{-5} , batch 128, 150 epochs), and **accuracy** as the headline metric. We restrict to the two datasets the genomap paper uses for *classification* — Tabula Muris cell recognition and the five-dataset pancreas integration; the capsule’s other matrices (sea-squirt developmental stages, retinal-bipolar) were used in the paper for trajectory and clustering/visualization, not classification, so they are excluded here.

Table 5: Single-cell recognition vs. the genomap paper. Test accuracy of our three model variants under the genomap paper’s protocol, with genomap’s own reported genoNet accuracy for reference (Tabula Muris 93%, Suppl. Fig. S4; Segerstolpe pancreas 97%, Fig. 7c). genoNet’s 2D-image CNN is the stronger classifier; our general transformer transfers unchanged from bulk to single-cell and lands within striking distance at far fewer parameters. Best per row (among our variants) in **bold**.

Task	Dataset (cells / cls)	Our model (accuracy)			genomap (paper)
		Independent	Shared	MoR	
Cell recognition	Tabula Muris (54,865 / 55)	0.830	0.829	0.826	0.93
Data integration	Pancreas (14,767 / 15)	0.935	0.920	0.904	0.97

Notes. Accuracy on the paper’s held-out test set (single split, seed 42). genomap’s genoNet reaches 93% on Tabula Muris (Suppl. Fig. S4; 6% / 21% above Cell-ID / SingleR) and 97% on the Segerstolpe pancreas subset (Fig. 7c). genoNet, a CNN purpose-built for the 2D genomap image, is stronger; our value is that the *same* 0.7M-parameter recursive transformer used for the TCGA cohorts (Table 1) transfers to single-cell unchanged. Source: `results_singlecell_paper/*/`.